P2: twist-pure coset selection for 2-power factor bases on prime-order curves

research/split_prime_ecdlp_20261107/followups/p2_twist_coset · run-20261107T214118Z · 2026-11-07

Outcome (frozen rule 4): toy-scale gain confirmed at stage level, smaller than first written [pcc:unverified]. F4 cost on planted targets is coset-independent; exact coverage follows c_E^m. The frozen statistic $(c_{\mathrm{sel}}/c_{\mathrm{rand}})^m \geq 2.25$ at $t = 4$; after the red-team correction the gain that matters for an index calculus is $(c_{\max}/c_{\mathrm{mean}})^{m-1}$, about 1.9 ($m = 2$) and 3.5 ($m = 3$) at $t = 4$, decaying toward 1 as t grows. A transfer of the lab's frame-choice lever to prime-order curves, not a new mechanism. S, end-to-end cost and any rho comparison are unset. Card `CARD-p2-twist-coset-20261107-1`, director assessment: executed as specified. Host: this container, L0; only exact counts decide.

1. Question, instances, method

A 2-power base V_c (Kummer $x^{2^t} = c$; Dickson torus $D_{2^t}(x) = c$) has about half its 2^t elements on E, half on the twist. Protocol (frozen 535d9da826, Addendum A, Clarification A.1, merged in #3657) asks whether F4 cost depends on c, whether exact coverage scales as c_E^m, and whether picking the coset with the largest c_E among C candidates pays. Primes: Kummer 1,049,089 and 16,777,729; Dickson 1,049,599 and 16,780,031; three prime-order curves each; $t \in \{4, 6, 8\}$; $C = \min(10^4, \mathrm{cosets})$; msolve 0.6.5 with 1.5 GiB and 1,200 s per job. Instrument `p2coset` (Rust); analysis `p2analyze` (Rust) applies the frozen rules to the raw JSONL.

2. Results

rule	test	result
1 lemma application	nontrivial E-target relations with twist part $\neq O$; exhaustive S_3 at 5 primes $\leq 10^4$, S_4 at 101, 151, 199; 3 curves each	0 violations in 24 runs
2 twist columns inert	F4 work, D, widest: selected vs random coset, 12 groups × 4 targets	max rel. diff 0.39 % (A/A 0.45 %); D and widest identical; kill line 10 %
3 coverage law	coverage ratio vs $(c_{\mathrm{sel}}/c_{\mathrm{rand}})^2$, 36 cells	max deviation 0.93 %; 0 cells beyond 20 %
4 gain	$(c_{\mathrm{sel}}/c_{\mathrm{rand}})^m$ at $t = 4$, curve 0, 8 groups	2.25 to 125

M1: observed $c_{\max}/c_{\mathrm{mean}}$ within 10 % of the exact binomial order statistic in all 36 cells; means 1.878 / 1.458 / 1.225 (exact 1.886 / 1.471 / 1.230) at t = 4 / 6 / 8. Expected gain against a typical coset, $(c_{\max}/c_{\mathrm{mean}})^m$: about 3.5 and 6.6 at $t = 4$, 2.1 and 3.1 at $t = 6$, 1.5 and 1.8 at $t = 8$ ($m = 2, 3$). The rule-4 maximum (125) comes from one seed-drawn coset with only 3 E-columns of 16.

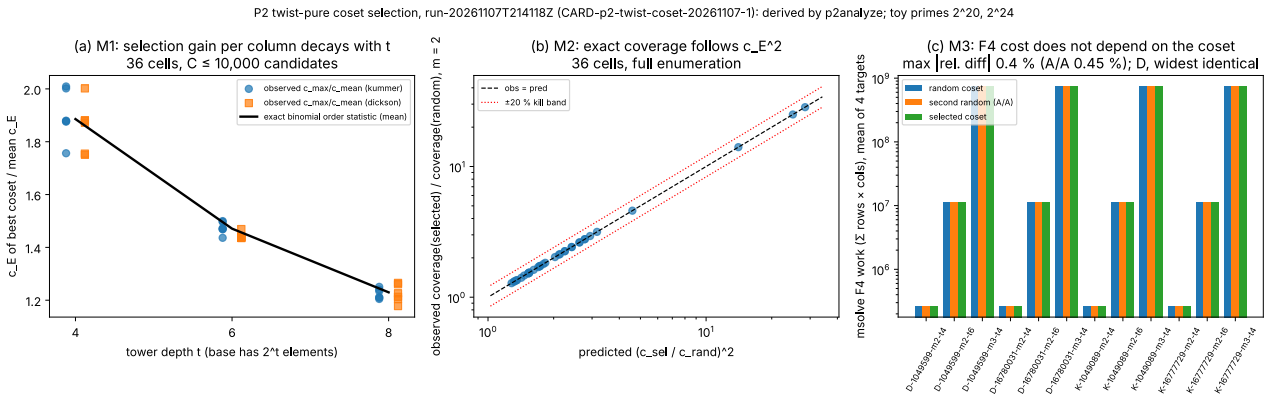


Figure 1: (a) order statistic vs t; (b) coverage law, 36 cells; (c) F4 work per arm, mean of 4 targets. Values derived by `p2analyze` from run-20261107T214118Z.

3. Red-team corrections

(1) Exponent $m-1$: an index calculus needs about c_E relations, so the net gain is $(c_{\mathrm{sel}}/c_{\mathrm{ref}})^{m-1}$; against the frozen random arm 1.5 to 25, against a typical coset about 1.9 / 3.5 ($t=4$), 1.46 / 2.1 ($t=6$), 1.22 / 1.5 ($t=8$). (2) A single seed-drawn random coset is a high-variance reference; c_{mean} is used instead. (3) F4 cost was measured on planted targets only; unplanted targets are untested. (4) The T* values for $m=3$ in the derived table use $m=2$ coverage and are wrong. Full review: `followups/red-team-p1-p2.md`.

4. Scope and caveats

Stage improvement of 2-power-base relation search on prime-order curves at 2^{20} and 2^{24} [pcc:unverified]; it changes no exponent ($c_{\max}/c_{\mathrm{mean}} \to 1$ as 2^t grows) and moves no leaderboard row (validation gate not run). Selection cost ($C \cdot 2^t$ Legendre symbols, about $3.8 \cdot 10^6$ field multiplications at $t=4$) is repaid before the first relation at these sizes (derived; mixes field multiplications with F4 mul-adds). The card cannot bind the binary it builds, so a post-run lint flags the binary-calling commands; the run recorded the binary hash. The relay added only the `BENCHREC=` prefix required by the repository hook. Purity is formalized abstractly as `twist_part_sum_eq_zero` (aburan28/crypto-lean#3).